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SECTION 01

PHASE 1: PROPOSAL

Strategic Real Estate Investment Intelligence: A Multi-Dimensional Analytics Solution for Property Valuation and Market Opportunity Mapping

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1.0 Problem Background

1.1 Business Context

The United States residential real estate market is an incredibly vast and complex ecosystem, currently valued at approximately \$43.1 trillion. For a modern real estate investment firm especially one managing a diversified portfolio of \$50 million or more, the primary challenge is no longer a lack of data, but the paradox of choice. With over 2.2 million active listings across the country, investment analysts are overwhelmed by raw information that is often siloed and disconnected.

Currently, most firms operate in a fragmented manner. They look at property listing prices on one screen and local economic news on another. This lack of a unified Business Intelligence (BI) system means that the firm's business context is reactive rather than proactive. The firm needs a way to automatically filter the entire national market to find "Goldilocks" zones which are the areas where the houses are high-quality, the local population is growing, and the prices are still low relative to the average household income.

1.2 Importance of the problem

The importance of solving this data fragmentation cannot be overstated, as the financial and operational risks of the status quo are severe:

- **Equity Erosion through Overpayment:** In real estate, profit is made at the buy. If a firm overpays for a property by just 10% due to poor market analysis, they suffer an immediate loss in equity. On a \$5 million acquisition, this is a \$500,000 mistake. Without a BI solution to calculate market comparables instantly, these errors occur frequently.
- **Information Asymmetry:** Real estate is hyper-local. Within a single county, prices per square foot can vary by over 200%. Traditional manual research cannot keep up with these micro-fluctuations. By the time a human analyst identifies a trend in a specific zip code, professional high-frequency investors have usually already bought the best assets.

- **Economic Misalignment:** A property does not exist in a vacuum. Its value is tied to the median household income of the city. If property prices in a city are rising while incomes are falling, a "bubble" is forming. Without integrating demographic data, an investment firm is flying blind, unable to distinguish between a growth market and a risky bubble.

1.3 Expected Impact

The implementation of this BI solution will move the firm from intuition-based to evidence-based decision-making. The expected impacts are categorized into three measurable areas. Firstly, Operational Efficiency. We expect a 99% reduction in research time. What currently takes an analyst team four weeks to research (50 zip codes) will be accomplished in 15 minutes through an interactive dashboard covering 15,000+ zip codes. Second, Financial Alpha (Outperformance). By identifying undervalued properties through the "Affordability Index" (Income vs. Price), the firm expects to increase its average annual return on investment (ROI) from a market-standard 8.5% to approximately 12.5%. For a \$50 million portfolio, this represents \$2,000,000 in additional annual revenue. Third, Risk Mitigation. By analyzing property condition and year built alongside price, the firm can avoid maintenance traps. This is expected to reduce annual portfolio volatility by 50%, ensuring more stable returns for stakeholders.

2.0 Objectives

1. To integrate multiple United States Housing datasets from different sources into a single clean and prepared dataset in Alteryx Designer.
2. To analyze factors affecting housing prices across the United States to find high opportunity investment zones.
3. 3. To create insightful BI dashboards using Power BI to analyze price trends, compare different housing demographics across cities in the United States for informed investment decisions.

3.0 Datasets Description

3.1 American House Prices

The first dataset was obtained from Kaggle, published by jeremylarcher. Titled American House Prices. It is a structured dataset consisting of 39,981 records and 14 variables.

Key Variables:

Variable	Description
Zip Code	Postal code of the listing.
Price	Listed house price in USD.
Beds	Number of bedrooms in the listing.
Baths	Number of bathrooms in the listing.
Living Space	Interior area in square feet.
Address	Street address of the listing.
City	City name where the listing is located.
State	State name where the listing is located.
Zip Code Population	Estimated number of individuals within the zip code.
Zip Code Density	Estimated population density per square mile
County	County where the listing is located.
Median Household Income	Median annual household income of the zip code area (USD).
Latitude	Latitude of the zip code.
Longitude	Longitude of the zip code.

3.2 USA House Prices

The second dataset was sourced from Kaggle, contributed by ahmedshahriarsakib, under the title USA House Prices. It is a structured dataset comprising 4,140 records and 18 variables.

Key Variables:

Variable	Description
Date	The date when the property was sold.

Price	The sale price of the property in USD.
Bedrooms	Number of bedrooms in the property.
Bathrooms	Number of bathrooms in the property.
Sqft Living	The size of the living area in square feet.
Sqft Lot	The size of the lot in square feet.
Floors	The number of floors in the property.
Waterfront	A binary indicator, 1 if the property has a waterfront view, 0 otherwise
View	An index from 0 to 4 indicating the quality of the property's view.
Condition	An index from 1 to 5 rating the condition of the property.
Sqft Above	The square footage of the property above the basement.
Sqft Basement	The square footage of the basement.
Yr Built	The year the property was built.
Yr Renovated	The year the property was last renovated.
Street	The street address of the property.
City	City name where the property is located.
Statezip	The state and zip code of the property.
Country	The country where the property is located.

3.3 USA Real Estate Dataset

The third dataset was obtained from Kaggle, published by fratzcan. Titled USA Real Estate Dataset. It is a structured dataset and the largest of the three, containing 2,226,382 records across 12 variables.

Key Variables:

Variable	Description
Brokered by	Broker or agency encoded.
Status	Status of the house whether it is for sale or sold

Price	House price in USD
Bed	Number of bedrooms in the house.
Bath	Number of bathrooms in the house.
Acre lot	Total land size or lot size in acres.
Street	The street address of the house.
City	City name where the house is located.
State	State name where the house is located.
Zip code	Zip code where the house is located.
House Size	House size or living space in square feet
Prev Sold Date	Previously sold date.

4.0 Proposed BI Approach

4.1 Technical Framework and Tool Utilization

The architecture of this Business Intelligence solution relies on the seamless orchestration between Alteryx Designer for data engineering and Power BI or Tableau for visual analytics. Alteryx will serve as the primary engine for the ETL (Extract, Transform, Load) process, converting raw, disparate CSV files from Kaggle into a high-performance, integrated data extract. The logic behind selecting Alteryx is its ability to create a repeatable and transparent workflow that ensures data lineage is maintained from the source to the final output. Once the data is refined, it will be transitioned to the visualization tool, which will serve as the front-end decision-support system. This tool will allow executive stakeholders to interact with the findings through dynamic filtering and drill-down capabilities, turning static data points into a narrative regarding market opportunities.

4.2 Data Preparation and Integration Workflow

The data integration process will be carried out in Alteryx Designer and follow ETL (Extract, Transform, Load) pipeline structure. The goal is to unify all three datasets into a single consolidated dataset that can be used for analysis and visualization in Power BI.

1. Data Integration

Three Input Data tools are used to load the datasets into the Alteryx workflow. Each dataset will retain its original column names and number of rows. Then, a Union tool will be used to merge the datasets by similar columns such as price, city, state and property size. Unique fields to each dataset are retained to preserve the richness of the dataset.

2. Data Cleaning

Each dataset is cleaned separately to ensure that the final dataset is consistent. Data records are dropped if critical data like price, city, state is missing to prevent it from affecting the analysis. String fields like city and state are preprocessed to remove any whitespace and standardized to a consistent format.

3. Data Transformation

Several transformations are applied to the data, after cleaning, to enhance the data for analysis. To allow comparisons between properties of varying size, the price per square foot field is calculated by dividing the property price by the size of the living area. An affordability ratio is the ratio between the property and the median household income of the area's zip code and is a critical indicator in determining whether a market will be undervalued or overpriced. Properties are also segmented by price tiers: Budget, Mid-Range, Premium, and Luxury to assist in segmentation analysis. All dates are presented in the same format to allow for time-series trend analysis. Average price, average price per square foot and total listings are also aggregated at the city and state level to create a geographic summary table which will feed the map and comparisons visualisation in the dashboard.

4.3 Planned Analysis and Visual Forms

This project will analyse the integrated dataset to identify the key factors influencing house prices across the United States. The analysis focuses on demographic conditions, geographic location, and property characteristics that affect property values. It also aims to understand housing affordability and market trends across different regions.

Six analyses are planned:

1. **Price vs Median Household Income Analysis using Bar Charts:** Compare average house prices against median household income across cities to identify markets where property values are significantly higher or lower relative to local earnings.
2. **Affordability Ratio Analysis using Bar Charts:** Derive an affordability ratio by dividing house price by median household income per zip code to rank cities and states from least to most affordable.
3. **Property Condition and Quality Analysis using Scatter Plots:** Examine how property condition rating, view score, and waterfront access correlate with sale price to determine which quality features most significantly drive property values.
4. **Property Age Analysis using Scatter Plots:** Assess how the year a property was built relates to its current sale price to determine whether newer properties consistently command higher market values compared to older stock.
5. **Demographic Influence Analysis using Bubble Charts:** Assess how population density and zip code population size relate to property prices, with bubble size representing total population to reveal the combined effect of demographic factors on housing costs.
6. **Lot Size and House Size Analysis using Scatter Plots:** Examine how lot size and interior house size together influence sale price across listings to determine the relative contribution of land and living space to property value.